

tcsBU - haplotype network visualization help

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Introduction

TCS ([Clement *et al.*, 2000](#)) is among the most popular haplotype network reconstruction methods. Written in Java, it is based on the *Statistical Parsimony* algorithm developed by [Templeton *et al.* \(1992\)](#). Unfortunately, TCS has some major drawbacks related with its output: 1) the graphics are coarse and often not ready for publication; 2) there is no built-in way of classifying haplotypes and displaying that information in haplogroups; and 3) when more than one network is present in the data, the software cannot use its 'spring' algorithm to layout multiple graphs, resulting in overlapping networks that have to be disentangled manually.

tcsBU (read as *TCS Bee You*) was developed to allow the production of publication-ready networks resulting from TCS analysis without too much effort. Users are able to:

- Classify haplotypes (for example, according to sampling locations or dates) and display this information as pie-chart-like haplogroups within the network.
- Visualize **several categorical and continuous traits at once** as **concentric rings** around each node — the group innermost, then each additional trait outward — with a matching multi-ring legend.
- Display haplotype names (e.g., H1, H2) on nodes when a three-column haplotype configuration file is provided, and toggle per-edge mutation-distance labels.
- Save the final network as **SVG (vector), PNG, JPG, or PDF** for publication or further editing.

This build. tcsBU here is the enhanced version embedded in **NetST**. When NetST drives it, the concentric multi-trait rings are configured on NetST's *Metadata* page and the configuration files below are generated automatically — you normally do not edit them by hand. The toolbar and dialogs described in this document match the current build (for example, trait rings replace the old single "Style"/"Color" toggles, and image export offers SVG/PNG/JPG/PDF).

The software

Dependencies

tcsBU is a browser-based JavaScript program that can be served by an HTTP server or run directly from a local folder. All computations are performed by the browser. **tcsBU** depends on the following third-party libraries (license type in parenthesis):

- [jQuery](#) (MIT): the underlying Javascript engine
- [w2ui](#) (MIT): JQuery-based User Interface
- [d3.js](#) (BSD): graphics engine (force-directed graph layout)
- [FileSaver.js](#) (X11/MIT): implements file saving in the browser
- [d3-svg-legend](#) (MIT): a modified version integrated into **tcsBU** source code

tcsBU requires JavaScript, SVG, FileReader, and Blob/FileSaver support. Use the Qt WebEngine bundled with NetST or a current desktop browser for standalone use.

Installation

For a web-based installation, save the **tcsBU** package into any directory under the webserver ROOT directory. No further steps are necessary. The only files/folders required are:

- index.html
- help.html
- tcsBU.js
- tcsBU.min.css
- jquery-3.2.1.min.js
- w2ui.min.js / w2ui.min.css
- d3.min.js

- FileSaver.min.js

If no web server is available, **tcsBU** can be saved into any local directory and opened directly in a browser using the file:// protocol.

Workflow overview

The typical workflow in **tcsBU** follows four sequential steps:

Step 1: Load a GML network file produced by TCS (*Load Data* button) → the haplotype network is drawn.

Step 2: Load a group configuration file from the *Groups* tab → classification groups (e.g., geographic locations) with their colors and patterns are defined.

Step 3: Load a haplotype configuration file from the *Haplotypes* tab → each sequence is assigned to a group; optionally, haplotype names (H1, H2, ...) are assigned to nodes.

Step 4 (optional): Add one or more traits → each haplotype node gains *concentric rings* (categorical rings segmented by class proportions, continuous rings coloured by a gradient). In **NetST** these rings are configured on the *Metadata* page and pushed to tcsBU automatically; a standalone trait configuration file can also be provided for a single continuous trait.

Important: Always load or define groups (Step 2) *before* loading haplotype classifications (Step 3), so that group names can be matched correctly.

Usage

Step 1: Loading a GML file

Click the **Load Data** button (top-left of the toolbar) to open a file browser. Select a tcsBU-compatible GML file. Classic TCS files may use *.graph*; NetST writes *.gml*. The files are usually found in the directory where the alignment used by TCS was stored. This action calls the internal loadGraph function, which parses the file, builds the internal node and edge lists, and starts the force-directed layout algorithm to display the raw haplotype network.

The GML (*Graph Modeling Language*) format is a plain-text format for network data. The first few lines of a typical TCS GML file look like this:

```
graph [
  directed 0
  node [
    data [
      Weight "outgroup weight= 0.01"
      Frequency "frequency=3
Seq1
Seq4
Seq5
"
      Sequence "Sequence =
```

```

AAA
"
    ]
    id 0
    label "Seq1"
    graphics [
.
.
.

```

This file should not be modified unless strictly necessary. The parser relies on the formatting and absolute position of certain keywords, so any editing must preserve the original indentation.

Haplotype labels in TCS files

For safe SVG and grid identifiers, tcsBU normalizes each GML member label: non-alphanumeric characters other than underscore become underscores, and a label beginning with a digit receives an **L** prefix. Configuration-file names pass through the same normalization. The safest practice is to keep original sample names unique after this transformation and to use the exact original names in the configuration files.

Step 2: Defining classification groups

Groups are managed in the **Groups** tab on the left panel. After loading a GML file, a 'Default' group (white, no pattern) is always present and cannot be modified or deleted. Any haplotypes not explicitly assigned to a group will belong to the Default group.

New groups can be added or deleted using the toolbar buttons within the Groups tab. Group names can be edited by double-clicking the name cell. **No two groups may share the same name**, as names are used as primary keys internally.

Each group (except Default) can be assigned a color and a fill pattern. Colors are specified in hexadecimal notation (e.g., #AF0026). Available pattern names are:

```

circles-1, circles-2,
lines-1, lines-2, lines-3, lines-4, lines-5, lines-6, lines-7, lines-8,
cross-1, cross-2, cross-3, cross-4

```

Click the **Load** button in the Groups tab toolbar to load a previously saved group configuration file (groupconf). This calls the internal loadGroups function. The Save button exports the current group list to a CSV file for later reuse. See [Group configuration file format](#) for details.

Step 3: Classifying haplotypes

After groups have been defined, click the **Load** button in the **Haplotypes** tab to load a haplotype configuration file (hapconf). This calls the internal loadHaplotypes function. The file assigns each sequence to a group and, optionally, gives a haplotype name to each node.

tcsBU supports two hapconf file formats, detected automatically by the number of columns:

- **Two columns — sequence name; group name.** Assigns each sequence to a group. Node labels remain the first sequence name in each node.
- **Three columns — sequence name; group name; haplotype name (for example, H1 or H2).** Assigns sequences to groups and supplies the display label for the existing node that contains the sequence. It does not regroup nodes or change topology. The **Haplotype** toolbar button is enabled only after a three-column file is loaded.

Sequence names must exactly match those in the original GML file (remember to prepend 'L' where names started with a digit). Group names must match those defined in the Groups tab. See [Haplotype configuration file format](#) for examples.

Step 4: Trait data and concentric rings

Beyond the primary group, additional traits can be layered onto each node as **concentric rings**. There are two paths, depending on how tcsBU is used:

- **In NetST (recommended):** configure any number of categorical and continuous traits on NetST's *Metadata* page — set each trait's type, whether it is visualized, and its colours — then click **Apply Visualization Config**. NetST rebuilds the ring configuration and refreshes tcsBU without reconstructing the network. This uses tcsBU's internal loadMetaConfig path.
- **Standalone single trait:** a single continuous trait can be supplied as a two-column, semicolon-delimited file (sequence name; numeric value, no header) via the internal loadTraits path. See [Trait configuration file format](#).

For a continuous trait, tcsBU determines the minimum and maximum values across matched sequences and maps each value to a position on that trait's colour gradient with a linear scale. For a categorical trait, each ring is segmented by the class proportions of the node's members. A legend showing every class swatch and each continuous gradient's range is displayed when the **Legend** is toggled on.

Node rendering: the concentric-ring model

Each haplotype is drawn as a circle whose radius scales with its frequency (the number of sequences it represents). Trait information is layered as concentric rings, ordered from the centre outward:

- In a NetST metadata configuration, the designated **Group** (primary categorical trait) occupies the **innermost** ring, drawn as a pie chart coloured by group.
- Each additional enabled trait is drawn as a further ring moving **outward**, in the order defined on NetST's Metadata page. A **categorical** ring is segmented by the class proportions of the node's members; a **continuous** ring is segmented by member value and coloured by that trait's own range and gradient.

Ring thickness and the node/edge/ring line widths are adjustable in the [Advanced settings dialog](#). The tcsBU **Legend** lists all rings in the same inner-to-outer order.

Note: This build uses a single default categorical palette and a single default numeric gradient; the older per-node "Style" and gradient "Color" toolbar toggles have been removed. Colours are set per trait (in NetST, on the Metadata page).

Force-directed layout algorithm

Once a GML file is loaded, the force-directed layout algorithm automatically starts and positions the nodes. The algorithm models haplogroups as electrically charged particles that repel each other, while edges act as springs that hold connected nodes together. All layout parameters live in the **Force-Directed Layout Settings** section of the [Advanced settings dialog](#) (toolbar **Advanced** button):

- **Link Distance:** the target spring length of edges, stretching or compressing the whole network.
- **Link Strength [0–1]:** how rigidly edges hold their target distance.
- **Friction [0–1]:** velocity decay per iteration. A value of 1 is frictionless; 0 freezes all nodes.
- **Charge:** electrostatic charge of each node (scales with node size). Negative values cause repulsion (recommended). The more negative, the stronger the repulsion.
- **Gravity [0–1]:** a weak spring pulling each node toward the center of the canvas, preventing the network from drifting outside the visible area.

Use the **Start / Stop** buttons in the same section to re-run or freeze the simulation after changing parameters. Default values work well for most data sets; avoid adjusting them unless necessary.

Advanced settings dialog

The toolbar **Advanced** button opens a settings dialog that can be **moved by dragging its title bar**. It is organized into three sections:

Force-Directed Layout Settings

Link Distance, Link Strength, Friction, Charge, Gravity, plus **Start / Stop** to re-run or freeze the layout (see above).

Node and Edge Settings

- **Node Radius** — radius (px) of a frequency-1 node; relative sizes of higher-frequency nodes are preserved when this changes.
- **Node Line Width** — stroke width of the node outline.
- **Edge Line Width** — base stroke width of the edges (links) between nodes.
- **Edge Weight Scale** — maximum multiplier applied to Edge Line Width when the toolbar's Edge Weight toggle is on. Edges approach the base width as their Changes value increases.
- **Text Offset** — gap between a node and its label text.

- **Haplotype Font Size** and **Name/ID Font Size** — global label sizes; the Info panel can override the position and size of an individual node's labels.

Metadata Ring Settings

- **Ring Line Width** — stroke width drawn between the segments of each trait ring.
- **Ring Width Ratio** — each outer (non-group) ring's width as a fraction of the node radius; the actual thickness is $\text{node radius} \times \text{ratio} \times \text{the ring's outer-ring scale}$.
- **Outer Ring Ratios** — comma-separated thickness ratios, from the innermost outer ring to the outermost (missing values default to 1).

Enter values and click **Apply** to redraw the network, or **Cancel** to close without changes. Inputs are validated (for example, widths must be zero or positive); invalid entries show an inline message.

Editing the network

Once a network is displayed, it can be explored and edited interactively using the toolbar buttons:

Button	Function
Load Data	Opens a file browser to load a GML network file (see Step 1). In NetST the network is loaded automatically, so this is mainly for standalone use.
Save Image (menu)	Exports the whole network as SVG (vector) , PNG , JPG , or PDF ; a system save dialog then chooses the location. See Saving the network .
Zoom In / Zoom Out	Scales the view in or out. The mouse wheel can also be used for zooming; clicking and dragging pans the view.
Delete Node (toggle)	When toggled on (button turns yellow), clicking any node deletes it from the network. Hovering over a node turns it red as a preview.
Delete Link (toggle)	When toggled on, clicking any edge (link) deletes it. Hovering over a link turns it red as a preview.
Legend (toggle)	Shows or hides the legend. The legend is draggable and can be repositioned anywhere on the canvas. Hover over it and use the mouse wheel to resize it; double-click it to restore the original size. It lists every trait ring — categorical

Button	Function
	class swatches and continuous-gradient ranges — in inner-to-outer order.
Haplotype (toggle)	Displays the haplotype name (e.g., H1, H2) next to each node. Enabled only after a 3-column hapconf file has been loaded.
Name/ID (toggle)	Shows or hides one member name/ID for each sampled haplotype node. Select a specific member and adjust its text position or size from the right Info panel.
Distance (toggle)	Shows or hides the mutation-distance (number of changes) label on each edge.
Edge Weight (toggle)	Weights edge thickness by mutation distance: fewer changes produce thicker edges. The base width and maximum multiplier are controlled by Edge Line Width and Edge Weight Scale in Advanced settings.
Undo	Restores the latest deleted node or link. Deletions are recorded in a stack, so Undo can be used repeatedly until the stack is empty.
Advanced	Opens the draggable Advanced settings dialog (force-directed layout; node and edge settings; metadata ring settings).

Individual nodes can also be dragged to any position by clicking and holding. Releasing a node will re-engage the force-directed simulation from that position. The left **Data** panel (Groups / Haplotypes) and the right **Info** panel can each be collapsed with the chevron buttons to give the network more room.

The right **Info** panel supports node search and shows details for a clicked node or edge. For nodes, it can display member sequences, group composition, metadata summaries, label controls, label X/Y positions and sizes, and highlighting. For edges, it shows endpoints and the changes count and can toggle highlighting or a per-edge changes label.

[Saving the network](#)

Use the **Save Image** toolbar menu to export the entire network. Four formats are offered:

- **SVG (vector)** — a standard image/svg+xml file. All parts of the network, including any nodes currently outside the visible area, are included. It can be embedded directly in a modern word processor or edited in vector software such as Inkscape or Adobe Illustrator®.
- **PNG** and **JPG** — raster images, exported at 2× resolution by default for sharper output.
- **PDF** — available when tcsBU is embedded in NetST; NetST converts the current serialized SVG to a vector PDF.

After choosing a format, a system save dialog lets you pick the location. The exported image preserves the legend's current position and size.

File formats

Group configuration file (groupconf)

A semicolon-delimited text file with no header row. Each line defines one group:

GroupName;#HexColor;PatternName

Color and pattern are optional. Examples:

```
Spain;#0af5d3;circles-1
Morocco South;#aea4b6
Morocco North;#54a1f2
Italy;#7684fa
Sardinia;#2425aa;lines-1
Portugal;#af0fff
France;#aaaa15
```

- Each line must have at least a group name.
- Colors must be in hexadecimal format (#000 000–#ffffff); invalid colors are silently ignored.
- Pattern names must match one of the allowed values listed in [Step 2](#); unknown pattern names are silently ignored.

Haplotype configuration file (hapconf)

A semicolon-delimited text file with no header row. Two formats are supported:

Two-column format (sequence name + group name):

```
L1543_IT-GIO;Italy
L1300_IT-SMR;Italy
L1860_MA-ZAO;Morocco
L2530_ES-IAL;Spain
L1821_ES-HER;Northern Spain
L1879_MA-RAS;Morocco
```

Three-column format (sequence name + group name + haplotype label):

L1543_IT-GIO;Italy;H1
L1300_IT-SMR;Italy;H1
L1860_MA-ZAO;Morocco;H2
L2530_ES-IAL;Spain;H3
L1821_ES-HER;Northern Spain;H4
L1879_MA-RAS;Morocco;H2

When a three-column file is loaded, column 3 supplies a display label for each existing sampled node, using the first sequence name in that node as the lookup key. The network topology and node membership come from the GML and are not changed by hapconf. The **Haplotype** toolbar button then becomes enabled.

Reminder: Sequence names must exactly match those in the GML file. If any name originally started with a digit, it will have been automatically prepended with 'L' by **tcsBU** — the same 'L' prefix must be used here. Group names must match those defined in the Groups tab.

Trait configuration file (traitconf)

A semicolon-delimited text file with no header row. Each line associates one sequence with a numeric trait value:

SequenceName;NumericValue

Example (collection year as trait):

L1543_IT-GIO;2005
L1300_IT-SMR;2008
L1860_MA-ZAO;2011
L2530_ES-IAL;2003
L1821_ES-HER;2015
L1879_MA-RAS;2009

- The numeric value can be an integer or decimal number.
- Sequences not present in the currently loaded network are silently ignored.
- A linear color scale maps the minimum value to one end of the gradient and the maximum value to the other.
- Within each node, the continuous-trait ring is subdivided into sectors proportional to the number of sequences at each distinct trait value.

References

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- Santos AM, Cabezas MP, Tavares AI, Xavier R, Branco M (2016) tcsBU: a tool to extend TCS network layout and visualization. *Bioinformatics* 32(4), 627–8. (doi: 10.1093/bioinformatics/btv636)